Frontend component specs and navigation rules

Reference implementation contracts; not deployed UI. Separate role authorization, paid entitlement, institutional module readiness and remaining quota.
allow = authenticated && resource_authorized && (free_core || effective_entitlement) && module_ready && quota_available.
Never grant grade/registrar/finance authority from a paid plan. Never require a personal upgrade to submit enrolled course work, view published course grades, obtain required institutional services or export required records.

Shell and route gates

WorkspaceShell: active institution/personal context, search, inbox, profile; capability projection from backend. Student/faculty/company/operator routes depend on current scoped permissions, not subscription names. BillingAccountSelector appears only for explicit account grants. FeatureGate distinguishes forbidden (no upsell), upgrade eligible, institution approval required, quota exhausted and backend unavailable.

Membership components

PlanComparison: Free/Plus/Pro cards with monthly/yearly toggle, total charge, renewal cadence, storage/AI limits, and sponsored status. Draft pricing must not be published before approval.
InstitutionOffering: buyer-only contact/quote flow, enrollment basis, minimum annual fee, implementation scope, security/privacy/accessibility review and authorized agreement.
CheckoutSummary: server-priced line items, currency/tax, cadence, next renewal, terms/cancellation acknowledgement; client cannot submit amount. PaymentMethodForm embeds/tokenizes through approved rail; never raw card storage in app.
CheckoutPending: verify backend payment/subscription status after return; no client-side unlock on URL parameters.
SubscriptionManager: current sponsor/personal plan, effective dates, pending changes, cancellation, invoice links, preserved quota budget sources.
UsageMeter: separate personal and institution-funded AI/storage, reset period, upcoming action credit cost; no silent sponsor budget consumption.
UpgradeDialog: display eligible feature and proposed plan; academic authorization failures never become a sales dialog.
BillingHistory: authorized invoice DTO; masked payment metadata; no internal payment tokens/diagnostics.
InstitutionActivationChecklist: signed agreement, covered institutions, authorized admin, DPA/security/accessibility prerequisites, SSO/roster readiness, migration acceptance and go-live signoff.

Education components

StudentToday: own due work, personal tasks, classes, service updates with source links and freshness.
CourseHome: authorized enrollment and published parent/item; no personal paid gate for required course access.
AssignmentSubmission: draft/upload/status/receipt; server checks deadline and idempotency; connectivity uncertainty triggers lookup before retry.
GradeView: own published result, distinguishes working and official grade; no paid access to peers/drafts.
FacultyStudio/Roster/ReviewQueue: assigned section, independent permissions, no global isFaculty flag.
PersonalNotes/Planner: owner scope, paid extras separate from core functions.

Company/ops components

SalesPipeline -> approved quote -> agreement -> rollout -> customer success; internal modules use business permissions, not student Pro.
DocumentReview -> exact version/hash, assigned reviewer, approval quorum, verified signatures, holds.
InvoiceConsole -> issue/credit/refund through approver commands; no arbitrary posted-value edits.
OperationsConsole -> scoped diagnostics; support access temporary and audited.

API behavior

Route handler file returns snake_case read projections intentionally; map to consistent frontend DTOs before broad deployment. All mutation requests carry idempotency UUID. Error envelope has code/message/requestId. 401 login, 403 unauthorized with no upsell, 404 unavailable object, 409 refresh/reconcile, 501 adapter absent, 503 retryable backend failure. Accessible error/status notifications and keyboard focus recovery. No optimistic activation of paid service.